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$$A = \begin{bmatrix} 5 & -3 & -2 \\ 8 & -5 & -4 \\ -4 & 3 & 3 \end{bmatrix}$$

$$\det(\lambda I_3 - A) = (\lambda - 1)^3 = 0$$

$$\Rightarrow \lambda = 1 \quad \text{w/ alg. multiplicity} = 3$$

$$\text{Solve } (A - \lambda I) v = 0$$

$$\Rightarrow v = c_1 \begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

$$\Rightarrow V_{\lambda=1} = \text{span} \left\{ \begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \right\}$$

$$\Rightarrow \text{geo. multiplicity of } \lambda = 1 \text{ is } 2$$

we want another generalized vector v_3 s.t.

$$A v_3 = 1 v_3 + (v_1 \text{ or } v_2 \text{ or } \dots)$$

want:

$$A v_1 = \lambda v_1$$

$$A v_2 = \lambda v_2$$

$$A v_3 = \lambda v_3 + v_2$$

$$(\lambda = 1) \quad \rightsquigarrow \begin{bmatrix} \boxed{\lambda} & & \\ & \boxed{\lambda} & \\ & & \boxed{\lambda} \end{bmatrix}$$

$$\text{choose } v_1 = \begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix}, \text{ and } v_2 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + \alpha \begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix}$$

$\leftarrow v_1$

$$(\alpha \in \mathbb{R} \text{ TBD})$$

then solve

$$A v_3 = \lambda v_3 + v_2$$

$$\text{i.e., } (A - \lambda I_3) v_3 = \begin{bmatrix} 3\alpha + 1 \\ 4\alpha \\ 2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4 & -3 & -2 \\ 8 & -6 & -4 \\ -4 & 3 & 2 \end{bmatrix} v_3 = \begin{bmatrix} 3\alpha + 1 \\ 4\alpha \\ 2 \end{bmatrix} \quad (*)$$

By Gauss-Jordan elimination

$$\left[\begin{array}{ccc|c} 4 & -3 & -2 & 3\alpha + 1 \\ 8 & -6 & -4 & 4\alpha \\ -4 & 3 & 2 & 2 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|c} 4 & -3 & -2 & 3\alpha + 1 \\ 0 & 0 & 0 & -2\alpha - 2 \\ 0 & 0 & 0 & 3\alpha + 3 \end{array} \right]$$

Solvability requires that $-2\alpha - 2 = 3\alpha + 3 = 0 \Rightarrow \alpha = -1$

$$\boxed{A v = b \text{ is solvable} \Leftrightarrow b \in C(A)}$$

$$\Leftrightarrow b \perp x, \forall x \in \ker(A^T)$$

$$\ker(B^T) = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \right\}$$

If $(*)$ is solvable, then

$$\begin{bmatrix} 3\alpha + 1 \\ 4\alpha \\ 2 \end{bmatrix} \perp \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad \& \quad \begin{bmatrix} 3\alpha + 1 \\ 4\alpha \\ 2 \end{bmatrix} \perp \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

$$\Leftrightarrow 3\alpha + 1 + 2 = 0 \quad \& \quad -2(3\alpha + 1) + 4\alpha = 0$$

$$\Rightarrow \alpha = -1$$

$$\text{If } \alpha = -1, \text{ then } v_2 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + (-1) \begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ -4 \\ 2 \end{bmatrix}$$

then (2) can be solved, we can pick

$$v_3 = \begin{bmatrix} -1/2 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} Av_1 = 1v_1 \\ Av_2 = 1v_2 \\ Av_3 = 1v_3 + 1v_2 \end{cases}$$

$$\Rightarrow A \begin{bmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{bmatrix} = \begin{bmatrix} | & | & | \\ v_1 & v_2 & v_3 + v_2 \\ | & | & | \end{bmatrix}$$

$$= \begin{bmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} \boxed{1} & & \\ & \boxed{1} & \\ & & \boxed{1} \end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix} 3 & -2 & -1/2 \\ 4 & -4 & 0 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} \boxed{1} & & \\ & \boxed{1} & \\ & & \boxed{1} \end{bmatrix} \begin{bmatrix} 3 & -2 & -1/2 \\ 4 & -4 & 0 \\ 0 & 2 & 0 \end{bmatrix}^{-1}$$

Say $A = \begin{bmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} \lambda & 0 & 0 \\ & \lambda & 0 \\ & & \lambda \end{bmatrix} \begin{bmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{bmatrix}^{-1}$

then

$$\exp(tA) = \begin{bmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{bmatrix} e^{t \begin{bmatrix} \lambda & 0 & 0 \\ & \lambda & 0 \\ & & \lambda \end{bmatrix}} \begin{bmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{bmatrix}^{-1}$$

$$= \begin{bmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} e^{t\lambda} & te^{t\lambda} & \frac{t^2}{2}e^{t\lambda} \\ 0 & e^{t\lambda} & te^{t\lambda} \\ 0 & 0 & e^{t\lambda} \end{bmatrix} \begin{bmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{bmatrix}^{-1}$$

then the sol'n to the IVP $\dot{x} = Ax$ $x(0) = x_0$

is given by $x(t) = e^{tA} x(0)$

$$= \begin{bmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} e^{t\lambda} & te^{t\lambda} & \frac{t^2}{2}e^{t\lambda} \\ 0 & e^{t\lambda} & te^{t\lambda} \\ 0 & 0 & e^{t\lambda} \end{bmatrix} \underbrace{\begin{bmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{bmatrix}^{-1} \begin{bmatrix} x_1(0) \\ x_2(0) \\ x_3(0) \end{bmatrix}}_{\text{constant vector}}$$

The general sol'n to $\dot{x} = Ax$ is given $c \in \mathbb{R}^3$ by

$$x(t) = \begin{bmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} e^{t\lambda} & te^{t\lambda} & \frac{t^2}{2}e^{t\lambda} \\ 0 & e^{t\lambda} & te^{t\lambda} \\ 0 & 0 & e^{t\lambda} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

$$= \underbrace{\begin{bmatrix} e^{\lambda t} v_1 & te^{\lambda t} v_1 + e^{\lambda t} v_2 & \frac{t^2}{2} e^{\lambda t} v_1 + te^{\lambda t} v_2 + e^{\lambda t} v_3 \end{bmatrix}}_{\Psi(t)} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

$\Psi(t)$ fund. set of sol'ns

$$\begin{aligned}
&= c_1 e^{\lambda t} v_1 \\
&\quad + c_2 e^{\lambda t} (t v_1 + v_2) \\
&\quad + c_3 e^{\lambda t} \left(\frac{t^2}{2} v_1 + t v_2 + v_3 \right)
\end{aligned}$$

Let $A \in M_n(\mathbb{C})$, and $f: \mathbb{C} \rightarrow \mathbb{C}$ analytic fun

s.t. $f(z) = \sum_{n \geq 0} c_n z^n$, then we can define

$$f: M_n(\mathbb{C}) \rightarrow M_n(\mathbb{C})$$

$$A \mapsto f(A) = \sum_{n \geq 0} c_n A^n$$

Suppose A admits the Jordan decomposition $A = P J P^{-1}$

$$\text{then } f(A) = P f(J) P^{-1}$$

$$= P \begin{bmatrix} f(J_1) & & 0 \\ & \ddots & \\ 0 & & f(J_m) \end{bmatrix} P^{-1}$$

$$\text{Suppose } J_i = \begin{bmatrix} \lambda_i & 1 & & 0 \\ & \ddots & \ddots & \\ 0 & & \lambda_i & 1 \end{bmatrix}_{m_i \times m_i}$$

then

$$f(J_i) = \begin{bmatrix} f(\lambda_i) & f'(\lambda_i) & \frac{f''(\lambda_i)}{2!} & \dots & \frac{f^{(m_i-1)}(\lambda_i)}{(m_i-1)!} \\ & \ddots & \ddots & \ddots & \\ 0 & & \lambda_i & 1 & \\ & & & \ddots & \\ & & & & f(\lambda_i) \end{bmatrix}$$

Complex eigenvalues (Boyce & DiPrima, Sec. 7.6)

$$\dot{x} = \begin{pmatrix} -\frac{1}{2} & 1 \\ -1 & -\frac{1}{2} \end{pmatrix} x \quad (x(t) \in \mathbb{R}^2 \quad \forall t)$$

What is the general sol'n in **real** form?

$$\text{Calculate } \det \begin{pmatrix} -\frac{1}{2} - \lambda & 1 \\ -1 & -\frac{1}{2} - \lambda \end{pmatrix} = 0 \Rightarrow \lambda = -\frac{1}{2} \pm i$$

$$\text{Let } \lambda_1 = -\frac{1}{2} + i, \text{ the } A v_1 = \lambda_1 v_1 \Rightarrow v_1 = \begin{pmatrix} 1 \\ i \end{pmatrix}$$

$$\left(\text{Let } \lambda_2 = -\frac{1}{2} - i, \text{ the } A v_2 = \lambda_2 v_2 \Rightarrow v_2 = \begin{pmatrix} 1 \\ -i \end{pmatrix} \right)$$

Consider

$$e^{\lambda_1 t} v_1 = \begin{pmatrix} 1 \\ i \end{pmatrix} e^{(-\frac{1}{2} + i)t}$$

$$= \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + i \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] e^{-t/2} (\cos t + i \sin t)$$

$$= e^{-t/2} \begin{pmatrix} \cos t \\ -\sin t \end{pmatrix} + i e^{-t/2} \begin{pmatrix} \sin t \\ \cos t \end{pmatrix}$$

$\Rightarrow$ The general sol'n is given by

$$x(t) = c_1 e^{-t/2} \begin{pmatrix} \cos t \\ -\sin t \end{pmatrix} + c_2 e^{-t/2} \begin{pmatrix} \sin t \\ \cos t \end{pmatrix}$$

$c_1, c_2 \in \mathbb{R}$ arbitrary.

Consider

$$\frac{d^ny}{dt^n} + a_{n-1} \frac{d^{n-1}y}{dt^{n-1}} + \dots + a_1(t) \frac{dy}{dt} + a_0(t) y = f(t)$$

w/ $a_0, a_1, \dots, a_{n-1} : \underset{\substack{\uparrow \\ \text{open interval}}}{I} \subseteq \mathbb{R} \rightarrow \mathbb{R}$ continuous function

Let $x_1 = y$

$$x_2 = \frac{dy}{dt} = y'$$

$\vdots$

$$x_n = \frac{d^{n-1}y}{dt^{n-1}} = y^{(n-1)}$$

$$\& \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$\frac{dx}{dt} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} x'_1 \\ x'_2 \\ \vdots \\ y^{(n-1)} \\ y^{(n)} \end{bmatrix}$$

$$= \begin{bmatrix} x_2 \\ x_3 \\ \vdots \\ x_n \\ -a_{n-1}(t)x_n - \dots - a_1(t)x_2 - a_0(t)x_1 + f(t) \end{bmatrix}$$

$$= \underbrace{\begin{bmatrix} 0 & 1 & & & \\ & 0 & 1 & & \\ & & \ddots & \ddots & \\ & & & 0 & 1 \\ -a_0(t) & -a_1(t) & \dots & -a_{n-1}(t) & 0 \end{bmatrix}}_{A(t) \in M_n(\mathbb{R})} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}}_x + \underbrace{\begin{bmatrix} 0 \\ \vdots \\ 0 \\ f(t) \end{bmatrix}}_{g(t)}$$

in the form $\frac{dx}{dt} = A(t)x + g(t)$

- homogeneous if $g(t) \equiv 0$
- inhomogeneous if $g(t) \not\equiv 0$

Suppose

$$\frac{d^ny}{dt^n} + A_{n-1}(t) \frac{d^{n-1}y}{dt^{n-1}} + \dots + A_1(t) \frac{dy}{dt} + A_0(t) y = f(t)$$

$$y(t) \in \mathbb{R}^k, A_i(t) \in M_k(\mathbb{R})$$

we can similarly get

$$\frac{dx}{dt} = \begin{bmatrix} 0 & I_k & & & \\ & 0 & I_k & & \\ & & \ddots & \ddots & \\ & & & 0 & I_k \\ -A_0(t) & -A_1(t) & \dots & -A_{n-1}(t) & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ \vdots \\ 0 \\ f(t) \end{bmatrix}$$

$$\text{w/ } x(t) \in \mathbb{R}^{nk}, \quad x(t) = \begin{bmatrix} y(t) \\ y'(t) \\ \vdots \\ y^{(n-1)}(t) \end{bmatrix}.$$

Consider the ode $\dot{x} = A(t)x$, $x(t) \in \mathbb{R}^n \forall t$

Observe that we can define a linear operator

$$L: \begin{matrix} C^1\text{-functions} \\ \mathbb{R} \rightarrow \mathbb{R}^n \end{matrix} \longrightarrow \begin{matrix} C^0\text{-functions} \\ \mathbb{R} \rightarrow \mathbb{R}^n \end{matrix}$$

$$f \mapsto \frac{df}{dt} - A(t)f$$

then the solution space of $\dot{x} = A(t)x$ is given by $\ker L$, which is a subspace.

Claim $\mathbb{R}^n$ is isomorphic to $\ker L$

define $f: \mathbb{R}^n \longrightarrow \ker L$
 η
 space of
 initial
 conditions

$$\eta \mapsto x(t; \eta)$$

where $x(t; \eta)$ is sol'n to IVP $\dot{x} = A(t)x$, $x(0) = \eta$

It's easy to check that f is linear.

Next we verify that f is bijective.

• inj: Indeed, if $x(t) \equiv 0 \Rightarrow x(0) = 0$

• Surj: Indeed, $\forall x(t) \in \ker L$, take $x(0) = \eta$

we need to verify that f is well-defined

$\left(f: A \rightarrow B \text{ is well-defined if } \forall x \in A, \exists! y \in B, \text{ s.t. } y = f(x) \right)$

By existence & uniqueness theorem, f is well-defined.

Or we can define

$$g: \ker L \longrightarrow \mathbb{R}^n$$

$$x(t) \longmapsto x(0)$$

again it's easy to check g is linear
we verify that g is bijective.

• Inj: it suffices to show $\ker g = \{0\}$. Indeed,

if $x(0) = 0$, then $x(t) = 0 \ \forall t$

• Surj: by existence & uniqueness thm,

$\forall x(0) \in \mathbb{R}^n$, $\exists!$ $x(\cdot) \in \ker L$.

□